

5.1.5 Plant and animal responses

Plant responses to environmental changes are coordinated by hormones, some of which are important commercially.

In animals, responding to changes in the environment is a complex and continuous process, involving nervous, hormonal and muscular coordination.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) the types of plant responses (ii) practical investigations into phototropism and geotropism	Plant response to abiotic stress and herbivory. Herbivory response to include chemical defences (alkaloids and pheromones only) and folding in response to touch AND the range of tropisms in plants. <i>M1.3, M1.6</i> PAG11 HSW4
(b) the roles of plant hormones	To include the role of hormones in leaf loss in deciduous plants, seed germination and stomatal closure.
(c) the experimental evidence for the role of auxins in the control of apical dominance	To include the effects of auxin concentration on apical dominance. Learners should be able to apply their knowledge and understanding to different experiments. Learners are not required to recall specific experiments. HSW5
(d) the experimental evidence for the role of gibberellin in the control of stem elongation and seed germination	To include the effects of gibberellin concentration on stem elongation. Learners should be able to apply their knowledge and understanding to different experiments. Learners are not required to recall specific experiments. HSW5
(e) practical investigations into the effect of plant hormones on growth	An opportunity for serial dilution. <i>M0.2, M1.1, M1.2, M1.3, M1.4, M1.6, M1.9, M1.10, M3.1, M3.2</i> PAG11 HSW4
(f) the commercial use of plant hormones	To include the use of hormones to control ripening, the use of rooting powders and hormonal weed killers. HSW12

Learning outcomes	Additional guidance
(g) the organisation of the mammalian nervous system	To include the structural organisation of the nervous system into the central and peripheral systems AND the functional organisation into the somatic and autonomic nervous systems.
(h) the structure of the human brain and the functions of its parts	To include the gross structure of the human brain AND the functions of the cerebrum, cerebellum, medulla oblongata, hypothalamus and pituitary gland.
(i) reflex actions	To include knee jerk reflex with reference to the survival value of reflex actions. <i>M0.1, M0.2, M1.1, M1.2, M1.3, M1.6</i> PAG11 HSW4
(j) the coordination of responses by the nervous and endocrine systems	To include the 'fight or flight' response to environmental stimuli in mammals AND the action of hormones in cell signalling (studied in outline only) with reference to adrenaline (first messenger), activation of adenylyl cyclase, and cyclic AMP (second messenger).
(k) the effects of hormones and nervous mechanisms on heart rate	An opportunity to monitor physiological functions, for example with pulse rate measurements before, during and after exercise or sensors to record electrical activity in the heart. <i>M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.10, M3.1</i> PAG10, PAG11 HSW4
(l) (i) the structure of mammalian muscle and the mechanism of muscular contraction (ii) the examination of stained sections or photomicrographs of skeletal muscle.	To include the structural and functional differences between skeletal, involuntary and cardiac muscle AND the action of neuromuscular junctions AND the sliding filament model of muscular contraction and the role of ATP, and how the supply of ATP is maintained in muscles by creatine phosphate. PAG1, PAG10, PAG11 HSW4